HANZALA CHAUDARY

Chemical Engineering Undergraduate | University of Engineering & Technology, Lahore
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PROFESSIONAL SUMMARY

Chemical Engineering undergraduate at UET Lahore with industrial training in flexible packaging manufacturing and multilayer lamination at Packages Convertors Limited. Technical project experience includes shell-and-tube heat exchanger design, process calculations, equipment sizing, and cost estimation using Python and Excel. Familiar with Aspen Plus, MATLAB, and engineering design principles, with interests in process engineering, manufacturing, and industrial operations.

EDUCATION

University of Engineering & Technology (UET), Lahore *(2023 – Expected 2027)*

Bachelor of Science in Chemical Engineering *(7th Semester)*

- **Key Coursework:** Chemical Process Design • Unit Operations • Heat & Mass Transfer • Fluid Mechanics • Thermodynamics • Reaction Engineering • Process Safety & Hazard Analysis

PROFESSIONAL EXPERIENCE

Packages Convertors Limited — Production Intern, Lamination (July 2026 – September 2026)

- Analyzed operating principles and production workflows across **8 industrial machines**, examining process controls and multilayer lamination operations.
- Evaluated **4 critical process parameters**—coat weight, nip pressure, web tension, and line speed—and their influence on lamination quality and process stability.
- Compared **2 adhesive technologies**, solvent-based and solventless lamination, focusing on operating requirements and product quality considerations.
- Examined production changeovers, manual and automatic splicing, quality inspection procedures, and manufacturing practices involving **5S, WCM, Autonomous Maintenance, and HSE**.

ACADEMIC PROJECTS

Shell & Tube Heat Exchanger Design | Excel, Python & 3D Modeling

- Engineered a shell-and-tube heat exchanger using D.Q. Kern's method, integrating thermal sizing, pressure-drop calculations, and cost estimation in Excel and Python.
- Developed a 3D equipment model to visualize exchanger geometry and mechanical configuration.

Eco-Friendly Soap Manufacturing | Waste Cooking Oil

- Designed a batch saponification process converting waste cooking oil into soap, optimizing reactant ratios and evaluating product quality through 3 tests: pH, lather formation, and hardness.

Ceramic Slurry Storage Tank | Process & Mechanical Design

- Designed a ceramic slurry storage tank, performing 4 key engineering analyses: storage capacity, wall thickness, operating pressure, and material stress.

Career Awareness Community Project | Event Coordination

- Organized 2 career-awareness seminars for junior Chemical Engineering students, covering higher education, engineering career pathways, and industrial opportunities.

TECHNICAL SKILLS

Process Simulation & Design: Aspen Plus, PFD development, Equipment sizing
Engineering Software: MATLAB, SolidWorks, ALOHA
Programming & Data Analysis: Python, Microsoft Excel

CERTIFICATIONS

- Chemical Hazards and Process Safety Specialization *[Coursera | Oct 2026]*
- Microsoft 365 Fundamentals Specialization — Microsoft *[Coursera | July 2026]*